

PE Lab — Step-by-Step Build Manual (MATLAB R2024b)

All 5 experiments, every example, click by click. Exam: 1 hour, no figure given → you must build the circuit from the question text. Parameters below = manual values; in the exam, just replace the numbers.

A. All experiments in your manuals

Manual	Examples (you will build each one below)
Exp-01 Important Measurements on Signals	1.1 Single-phase half-wave diode rectifier · 1.2 Measurement with the Oscilloscope (Scope) block · 1.3 Measurement with the Multimeter block · 1.4 Measurement port · 1.5 Mean and RMS blocks · 1.6 Instantaneous power and average power · 1.7 Apparent power and power factor
Exp-02 Uncontrolled Rectifier Circuits	2.1 Goto and From blocks · 2.2 Making a subsystem · 2.3 Power block (1-φ and 3-φ) · 2.4 Freewheeling diode · 2.5 Disabling a block · 2.6 Fourier block · 2.7 Three-phase diode rectifier · 2.8 Power factor of a three-phase uncontrolled rectifier · 2.9 Measurement of conduction loss
Exp-03 Controlled Rectifier Circuits	3.1 Single-phase half-wave thyristor rectifier · 3.2 Single-phase full-wave thyristor rectifier (I) — 2 SCR + 2 diodes · 3.3 Full-wave thyristor rectifier (II) — 4 SCR · 3.4 Three-phase thyristor rectifier · 3.5 Effect of a filter capacitor on the rectifier · 3.6 Coupled inductors
Exp-04 DC-DC Converters	4.1 Buck converter · 4.2 Operating mode (CCM/DCM) · 4.3 Effect of input voltage changes · 4.4 Effect of output load changes · 4.5 Generation of PWM signals · 4.6 PWM Generator (DC-DC) block · 4.7 Closed-loop control of a buck converter · 4.8 Flyback converter · 4.9 Efficiency of a flyback converter
Exp-05 Inverters	5.1 Single-phase PWM inverter (unipolar) · 5.2 THD block · 5.3 Harmonic analysis with the FFT Analyzer · 5.4 Three-phase PWM inverter · 5.5 Connection Port block (subsystem with electrical ports)

B. Basic actions (learn these 8 once — every step below uses them)

Action	Exactly what to do
NEW	MATLAB command window → type <code>simulink</code> → Enter → click Blank Model . Immediately press <code>Ctrl+S</code> , name it (e.g. <code>exp1_hw</code>), Save.
ADD “X”	Single-click on an empty place of the canvas → start typing the block name X → a list appears → pick the entry whose path (grey text) contains Specialized Power Systems for electrical/measurement blocks, or Simulink for Scope, Display, Constant, Pulse Generator, Product, Gain, Mux, Demux, Goto, From, Step, Sum, Relational Operator, Logical Operator, Saturation, Transport Delay, Repeating Sequence, Sine Wave. Press Enter. If Quick Insert doesn't appear: Simulation tab → Library Browser → Simscape → Electrical → Specialized Power Systems → (Sources / Passives / Power Electronics / Sensors and Measurements / Control / Utilities); drag the block in.
SET	Double-click the block → type the value in the named box → OK . (MATLAB expressions like <code>1/60/4</code> , <code>220*sqrt(2)</code> , <code>2*pi*50</code> are allowed.)
WIRE A→B	Move the mouse onto a terminal of A until it becomes a crosshair → hold left button → drag to the terminal of B → release. To branch from an existing wire: hold Ctrl (or use right button) and drag from the wire. Electrical wires are plain lines; signal wires have arrows.
ROTATE / NAME	Click block → <code>Ctrl+R</code> (90° clockwise). <code>Ctrl+I</code> flips it. Rename: click the name text under the block, type a new name (e.g. “Load Voltage” for a Scope).
POWERGUI	ADD “powergui” → double-click → Solver tab → Simulation type = Discrete → Sample time (s) = <code>5e-6</code> → OK . (Without it: error “model must contain a powergui block”).
RUN	Simulation tab → Stop Time box → type value (e.g. <code>0.1</code>) → click Run (or <code>Ctrl+T</code>). Wait until the bottom-left says “Ready”.
SCOPE-READ	Double-click Scope. Autoscale: press the “Scale X & Y axes” button (or <code>Ctrl+A</code> in the scope). Zoom: Zoom-X button, drag over one steady-state cycle. Measurements tab (older look: Tools → Measurements): Cursor Measurements (ΔT , $1/\Delta T$ = frequency) and Signal Statistics (Max, Min, Peak-to-peak, Mean , Median, RMS — for the zoomed region).

Rules that prevent 90% of errors

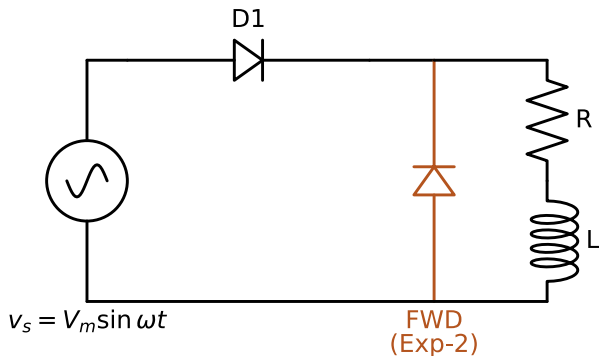
1. Every Diode and Thyristor: SET **Snubber resistance R_s (Ohms)** = `1e7`. Leave everything else default.
2. AC Voltage Source takes the **peak** amplitude: 220 V (rms) → `220*sqrt(2)`.
3. Mean / RMS / Fourier / THD blocks: take them from **Specialized Power Systems** → **Sensors and Measurements** (they have a “Fundamental frequency (Hz)” box). Set Sample time = same as powergui (`5e-6`) or leave the default.
4. Fundamental frequency of Mean/RMS = the repetition frequency of the waveform you measure: half-wave **f**, full-wave **2f**, 3- ϕ bridge **6f**, DC-DC **f_s** , inverter RMS/THD/Fourier **output f**.
5. Measurement ports (m) you don't use → ADD “Terminator” and wire to it (removes warnings).
6. Read Mean/RMS/Display values only after the circuit reaches steady state (last cycles). Display block: drag its corner to make it bigger if the number is cut.

1-hour exam plan (no figure) · 0–5 min: read, draw the circuit by hand on the answer script (use the schematics in this manual), write the theory value · 5–30 min: build + powergui + measurements · 30–40: run, read values, scope screenshots · 40–55: do the variation they ask ($\alpha / D / R / m_a$) and make a table · 55–60: save, write comparison “simulation vs theory” and the reason for the small difference (diode drop 0.8 V, R_{on}).

Exp-01 — Important Measurements on Signals

1.1 Single-phase half-wave rectifier (R load)

Question style: "Simulate a single-phase half-wave rectifier, $v_s = 311 \sin(2\pi \cdot 60t)$, $R = 10 \Omega$. Show load voltage."



Circuit in words: source → diode (anode on source side) → load → back to source. (Ignore the FWD for now.)

Block (ADD)	SET
AC Voltage Source	Peak amplitude 311 , Phase 0, Frequency 60
Diode	Snubber Rs 1e7
Series RLC Branch	Branch type R, Resistance 10
Voltage Measurement	—
Scope (rename "Load Voltage")	—
powergui	Discrete, 5e-6
Terminator	—

1. NEW → save as exp1_hw .
2. ADD all blocks in the table. SET each one as in the table.
3. Place the AC source on the left, Diode on top (horizontal, triangle pointing right), Series RLC Branch on the right. Select the RLC branch → Ctrl+R to make it vertical.
4. WIRE: AC source **top** terminal → Diode **left** terminal (anode a).
5. WIRE: Diode **right** terminal (cathode k) → RLC branch **top**.
6. WIRE: RLC branch **bottom** → AC source **bottom**.
7. Place Voltage Measurement to the right of the load. WIRE its + to the RLC top wire, its - to the RLC bottom wire.
8. WIRE: Voltage Measurement output v → Scope input.
9. WIRE: Diode **m** port → Terminator.
10. RUN with Stop Time 100e-3 → double-click Scope → autoscale.

Expected: positive half-sine humps, peak ≈ 310 V, zero in negative half cycles.

1.2 Measurement with the Scope block

1. Use the model of 1.1 (already run). Open the Scope.
2. Measurements tab → **Cursor Measurements**. Drag cursor 1 to the start of a hump and cursor 2 to the start of the next hump → read $1/\Delta T = 60$ Hz.
3. Turn cursors off. Click **Zoom X** → drag over exactly one full period (e.g. 0.05 s → 0.0667 s).
4. Measurements tab → **Signal Statistics**. Read Max, Min, Mean (= average), RMS.
5. For your answer script: scope window → File → **Copy to Clipboard** (or screenshot).

Manual: Max 310.2 V, Min ≈ 0 , **Mean 98.71 V**, **RMS 155.1 V**. Theory $V_m/\pi = 99.0$ V, $V_m/2 = 155.5$ V.

1.3 Measurement with the Multimeter block (parallel RLC, no source)

Circuit: $R = 10 \Omega$, $L = 250$ mH (initial current 0.1 A), $C = 2$ mF (initial voltage 2 V), all three in parallel. Show capacitor voltage and current.

1. NEW → save as exp1_multimeter .

2. ADD three "Series RLC Branch" blocks, rotate all vertical (Ctrl+R), rename them R, L, C.
3. SET R: Branch type R, Resistance 10 .
4. SET L: Branch type L, Inductance 250e-3 , tick **Set the initial inductor current**, Inductor initial current 0.1 .
5. SET C: Branch type C, Capacitance 2e-3 , tick **Set the initial capacitor voltage**, Capacitor initial voltage 2 ,
Measurements = Branch voltage and current.
6. WIRE all three tops together, and all three bottoms together (parallel).
7. ADD "Multimeter" (Sensors and Measurements). Double-click it → in the left list click Ib: C → >>; click Ub: C → >> → Close.
8. ADD "Demux" (Number of outputs 2) and two Scopes; rename the scopes "Capacitor Current" and "Capacitor Voltage".
9. WIRE Multimeter output → Demux input; Demux output 1 → Capacitor Current scope; output 2 → Capacitor Voltage scope (same order as the Multimeter list).
10. POWERGUI (Discrete 5e-6). RUN with Stop Time 0.5 .

Expected: damped oscillation starting at 2 V (voltage) decaying to 0. Current sign: current entering the terminal with "+" is positive.

1.4 Measurement port (diode current & voltage)

1. Open exp1_hw → File → Save As exp1_mport .
2. Delete the Terminator on the diode m-port.
3. ADD "Demux" (2 outputs) + two Scopes named "Diode Current" and "Diode Voltage".
4. WIRE Diode m → Demux; Demux out 1 → Diode Current; out 2 → Diode Voltage.
5. RUN (0.1 s). Use Signal Statistics on each.

Out 1 = i_{AK} (anode→cathode, peak ≈ 31 A); out 2 = v_{AK} (≈ 0.8 V when ON, down to -311 V when OFF → $PIV = V_m$). Use: diode rating = I_{avg} , I_{rms} , PIV.

1.5 Mean and RMS blocks — half-wave with RL load

$V_m = 100$ V, 60 Hz, $R = 100 \Omega$, $L = 0.1$ H. Find average and RMS of load voltage and current.

1. NEW → exp1_rl . ADD AC Voltage Source (Peak 100 , Freq 60), Diode (R_s 1e7), Series RLC Branch (Branch type **RL**, R 100 , L 0.1), Current Measurement, Voltage Measurement, powergui (Discrete 5e-6), Terminator.
2. WIRE: source top → Diode anode; Diode cathode → Current Measurement +; Current Measurement - → RL top; RL bottom → source bottom; Diode m → Terminator.
3. WIRE Voltage Measurement + → RL top, - → RL bottom.
4. ADD two "Mean" blocks and two "RMS" blocks (all from Sensors and Measurements). SET each: Fundamental frequency 60 .
5. ADD four "Display" blocks.
6. WIRE current output i → Mean → Display; branch the same i wire (Ctrl+drag) → RMS → Display. Do the same for voltage v.
7. Also: i → Scope "Current"; v → Scope "Load Voltage".
8. RUN with Stop Time 834e-3 (≈ 50 cycles, enough to reach steady state).
9. **Conduction angle**: open Load Voltage scope → zoom to one cycle → Cursor Measurements → put cursors at start and end of the negative part → ΔT . Angle = $\Delta T \times 60 \times 360$. Conduction angle = $180^\circ + \text{that angle}$.
10. (Optional, same scope) ADD "Gain" = 0.01 on v, ADD "Mux" (2 inputs): i → Mux in1, $v \times 0.01$ → Mux in2 → Scope "CurrentVoltage": you see current stays positive while voltage goes negative.

Manual: I_{avg} 0.3041 A, I_{rms} 0.4685 A, V_{avg} 30.4 V, V_{rms} 49.74 V; negative part 962.95 $\mu s = 20.8^\circ$ → conduction 200.8°.
Theory: $V_{avg} = (V_m/2\pi)(1 - \cos \beta) = 30.8$ V with $\beta = 200.7^\circ$.

1.6 Instantaneous power and average power

1. Save `exp1_rl` as `exp1_power`.
2. ADD "Product" (Simulink, Math Operations; 2 inputs by default).
3. WIRE (Ctrl+drag branches): $v \rightarrow$ Product in1; $i \rightarrow$ Product in2.
4. Product out $\rightarrow$ Scope "Instantaneous Power"; branch $\rightarrow$ Mean (Fundamental frequency 60) $\rightarrow$ Display "Average Power".
5. RUN (834e-3).

$P = 21.95 \text{ W}$ (check: $I_{\text{rms}}^2 R = 0.4685^2 \times 100 = 21.95 \text{ W}$ ✓). Power waveform frequency = 60 Hz.

1.7 Apparent power and power factor

1. Continue in `exp1_power`. ADD a second Voltage Measurement; WIRE its + to source top, - to source bottom (source voltage).
2. ADD RMS (Fundamental frequency 60) $\rightarrow$ gives $V_{\text{s,rms}}$. You already have I_{rms} from 1.5.
3. ADD Product: in1 = $V_{\text{s,rms}}$, in2 = I_{rms} $\rightarrow$ Display "Apparent Power".
4. ADD "Divide" (Simulink, Math Operations): top input ($\times$) = Average Power, bottom input ($\div$) = Apparent Power $\rightarrow$ Display "Power Factor".
5. RUN (834e-3).

$S = 33.13 \text{ VA}$, $\text{PF} = 0.6626$. $Q = \sqrt{S^2 - P^2} = 24.8 \text{ VAR}$.

Exp-02 — Uncontrolled (Diode) Rectifier Circuits

2.1 Goto and From blocks

1. Open `exp1_power` → Save As `exp2_goto`.
2. Delete the long signal wires from the Voltage Measurement output and Current Measurement output.
3. ADD “Goto” → SET Goto tag = `V`. WIRE load Voltage Measurement `v` → Goto [`V`].
4. ADD “Goto” → tag `I`. WIRE Current Measurement `i` → Goto [`I`].
5. ADD “From” → SET Goto tag = `V`; ADD another From → tag `I`. Wire each From output to the Mean/RMS/Product blocks that the old wires went to (copy a From block with Ctrl+C / Ctrl+V if you need more).
6. RUN — results must be identical to 1.7.

2.2 Making a subsystem

1. In `exp2_goto`: drag a rectangle with the mouse around the circuit only (source, diode, load, measurements, Goto blocks, Terminator).
2. Press `Ctrl+G` (or hover the “...” at the rectangle corner → Create Subsystem). Rename it “Rectifier”.
3. The From blocks turn red. Double-click the subsystem → double-click each Goto → **Tag visibility = global** → OK.
4. Click **Up to Parent** (arrow at the top of the canvas). From blocks are normal again.
5. (Optional) Delete the Display blocks, select all measurement blocks → `Ctrl+G` → name “Measurements” → open it, rename the Out ports (Average Power, Apparent Power, PF...) → Up to Parent → connect Displays to its outputs.
6. RUN — same values.

2.3 Power block (linear circuits only)

(a) Single-phase RL, no diode

1. NEW → `exp2_powerblock`. ADD AC Voltage Source (Peak `100`, Freq `60`), Series RLC Branch (RL, `100`, `0.1`), Current Measurement, Voltage Measurement, powergui (Discrete 5e-6).
2. WIRE: source top → Current Meas +; Current Meas - → RL top; RL bottom → source bottom. Voltage Meas across RL.
3. ADD “Power” (Sensors and Measurements; the single-phase one with inputs V and I) → SET Fundamental frequency `60`.
4. WIRE `v` → Power **V**; `i` → Power **I**. Outputs **P** → Display, **Q** → Display.
5. RUN (0.2 s).

$P = 43.78 \text{ W}$, $Q = 16.5 \text{ VAR}$ (inductive → positive Q). $PF = P/\sqrt{P^2+Q^2} = 0.936$. If you insert a diode, this block gives wrong values (10.94 W) → for rectifiers always use Product → Mean.

(b) Three-phase linear load

1. NEW → `exp2_power3ph`. ADD “Three-Phase Source” → SET Configuration **Yg**, Phase-to-phase voltage (Vrms) `207.8461`, Phase angle 0, Frequency `60`, untick Internal (impedance).
2. ADD “Three-Phase V-I Measurement” → Voltage measurement **phase-to-ground**, Current measurement **yes**, untick “in pu” boxes.
3. ADD “Three-Phase Series RLC Branch” → Branch type **RL**, R `10`, L `0.1`.
4. WIRE Source A,B,C → V-I Meas A,B,C; V-I Meas a,b,c → RLC branch A,B,C. WIRE the three right-side terminals of the RLC branch together (star point).
5. ADD “Power (3ph, Instantaneous)” → WIRE `Vabc` → `Vabc`, `Iabc` → `Iabc`. Outputs P, Q → Displays.
6. POWERGUI Discrete 5e-6. RUN (0.2 s).

$P = 284 \text{ W}$, $Q = 1071 \text{ VAR}$, $PF = 284/\sqrt{284^2+1071^2} = 0.2563$. $V_{\text{phase}} = 207.85/\sqrt{3} = 120 \text{ V rms}$.

2.4 Freewheeling diode

1. Open `exp1_rl` → Save As `exp2_fwd`.
2. ADD a second Diode (Rs `1e7`), rename "FWD", rotate so it is vertical with the triangle pointing **up** (anode at bottom, cathode at top): use `Ctrl+R` until the anode (a) is at the bottom.
3. WIRE FWD cathode → load top wire; FWD anode → load bottom wire (FWD in parallel with the RL load). Its m-port → Terminator.
4. RUN (834e-3). Read Mean of load voltage.

V_{avg} **31.25 V** (was 30.4 V); load voltage has no negative part. Theory $V_m/\pi = 31.83$ V.

2.5 Disabling a block

1. In `exp2_fwd`: right-click FWD → **Comment Out** (shortcut `Ctrl+Shift+X`). It turns grey.
2. RUN → V_{avg} goes back to 30.4 V.
3. Right-click → **Uncomment** to enable it again.

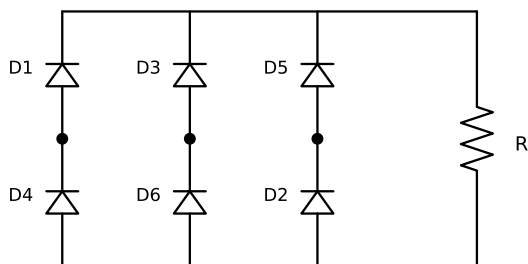
2.6 Fourier block (harmonics of load voltage)

1. In `exp2_fwd` (FWD enabled). ADD "Fourier" (Sensors and Measurements) → SET Fundamental frequency `60`, Harmonic n `0:5`, Initial input `[0, 0]`, Sample time `5e-6`.
2. WIRE load voltage v → Fourier input. Output **magnitude** → Display (rename "Magnitude"), output **phase** → Display "Phase (Degrees)". Enlarge the Displays (6 numbers each).
3. RUN (834e-3).

Magnitudes (peak): DC 31.83, n1 50.0, n2 21.22, n3 ≈ 0 , n4 4.244, n5 ≈ 0 V. (DC = V_m/π , fundamental = $V_m/2$.)

2.7 Three-phase diode (bridge) rectifier

3- ϕ , 220 V phase (rms), 50 Hz, $R = 10 \Omega$. Find average load voltage and power.



phase a,b,c → mid-points of legs 1,2,3; sources star-connected, star → Ground

Leg midpoints = phase a, b, c. Top diodes: anode at phase, cathode at + rail. Bottom: anode at - rail, cathode at phase.

Block	SET
AC Voltage Source x3 (V1, V2, V3)	Peak <code>220*sqrt(2)</code> , Freq <code>50</code> , Phase <code>0 / -120 / 120</code>
Ground	(Utilities)
Diode x6 (D1...D6)	Rs <code>1e7</code>
Series RLC Branch	R <code>10</code> , Measurements Branch voltage and current
Multimeter, Demux(2), Mean x2, Product, Displays, Scopes	Mean frequency <code>300</code> (= 6×50)
powergui	Discrete <code>1e-6</code>

Shortcut: instead of 6 diodes use ADD "Universal Bridge" → Number of bridge arms 3, Power Electronic device **Diodes**, Rs `1e7`, Cs `inf`. Its A, B, C go to the sources, + and - to the load.

1. NEW → `exp2_3ph_diode`. ADD all blocks. SET as the table.
2. Place V1, V2, V3 vertically side by side. WIRE all three **bottom** terminals together → Ground.
3. Place D1, D3, D5 in a row at the top (vertical, anode at bottom), D4, D6, D2 in a row below them (vertical, anode at bottom).
4. WIRE D1 anode → D4 cathode (leg a). Same: D3 anode → D6 cathode (leg b); D5 anode → D2 cathode (leg c).
5. WIRE V1 top → leg a midpoint; V2 top → leg b midpoint; V3 top → leg c midpoint.

6. WIRE D1, D3, D5 cathodes together = + rail; WIRE D4, D6, D2 anodes together = - rail.
7. Rotate R vertical: top → + rail, bottom → - rail.
8. All six m-ports → Terminators (or untick "Show measurement port" in each diode).
9. Multimeter: select Ub: ... and Ib: ... of the load branch → >>. Multimeter → Demux → out1, out2 (check order in the list: voltage/current).
10. Voltage → Mean(300) → Display "Average Load Voltage"; Voltage × Current → Product → Mean(300) → Display "Average Load Power"; voltage → Scope.
11. RUN with Stop Time 0.1.

$V_{avg} \approx 512.7 \text{ V}$ (theory $1.35 \times V_{LL} = 1.35 \times 381 = 514.4 \text{ V}$), $P \approx 26.33 \text{ kW}$. Ripple frequency = 300 Hz (6 humps per cycle).

2.8 Power factor of the three-phase rectifier

1. Save exp2_3ph_diode as exp2_3ph_pf.
2. Delete the wire V1 top → leg a. ADD Current Measurement: V1 top → +; - → leg a (current of phase a).
3. ADD Voltage Measurement: + → V1 top, - → V1 bottom (ground).
4. ADD RMS ×2 (frequency 50): $v_a \rightarrow \text{RMS}$, $i_a \rightarrow \text{RMS}$.
5. Product(v_a , i_a) → Mean (frequency 100) → Gain 3 → Display "P".
6. Product(V_{rms} , I_{rms}) → Gain 3 → Display "S".
7. Divide: $P \div S \rightarrow \text{Display "PF"}$. RUN (0.1 s).

$P = 26.42 \text{ kW}$, $S = 27.64 \text{ kVA}$, $\text{PF} = 0.9558$. $P_{\text{source}} - P_{\text{load}} = \text{diode loss}$; $\eta = 26.33/26.42 \approx 99.66\%$.

2.9 Conduction loss of a diode

1. In exp2_3ph_diode: remove D1's Terminator. D1 m → Demux(2) → out1 = i, out2 = v.
2. Product(i, v) → Scope "Diode loss" and → Mean (frequency 50) → Display.
3. RUN (0.1 s).

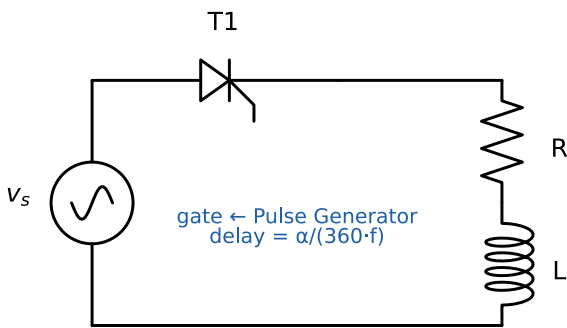
$\approx 15 \text{ W}$ per diode; loss is non-zero for 6.67 ms per 20 ms (one-third of the period — each diode conducts 120°).

Exp-03 — Controlled (Thyristor) Rectifier Circuits

Firing angle rule (Pulse Generator) · Pulse type *Time based* · Amplitude *1* · Period *1/f* · Pulse Width (% of period) *5* · Phase delay (secs) = $\alpha/(360 \cdot f)$ → type it directly, e.g. $\alpha = 60^\circ$, 50 Hz: $60/360/50$.

3.1 Single-phase half-wave thyristor rectifier (RL)

$V_m = 169.7 \text{ V}$, 60 Hz, $R = 10 \Omega$, $L = 10 \text{ mH}$, $\alpha = 90^\circ$.

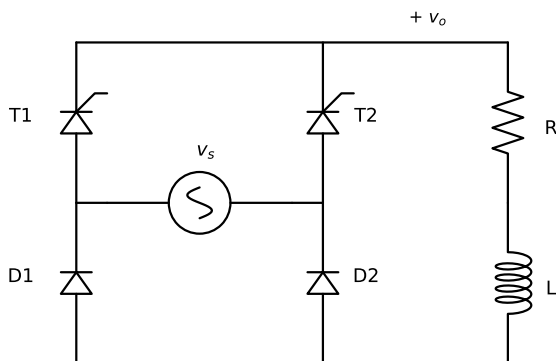


1. NEW → exp3_scr_hw . ADD AC Voltage Source (Peak 169.7 , Freq 60) , “Thyristor” (Specialized Power Systems → Power Electronics; Rs 1e7) , Series RLC Branch (RL, 10 , 10e-3) , Voltage Measurement, Scope, powergui (Discrete 5e-6), Terminator.
2. ADD “Pulse Generator” (Simulink Sources) → SET Time based, Amplitude 1 , Period 1/60 , Pulse Width 5 , Phase delay 1/60/4 (= 90°).

3. WIRE: source top → Thyristor **a** (anode); Thyristor **k** → RL top; RL bottom → source bottom.
4. WIRE Pulse Generator output → Thyristor **g** (gate). Thyristor **m** → Terminator.
5. Voltage Measurement across RL → Scope; also → Mean (60) → Display.
6. RUN (Stop Time 0.1).
7. Change α : Phase delay 1/60/12 (30°), 1/60/6 (60°) ... make a table of V_{avg} .

Output starts at α (90°), continues a little into negative half ($\beta \approx 200^\circ$). $V_{avg} \approx (V_m/2\pi)(\cos \alpha - \cos \beta)$: 90° → $\approx 25 \text{ V}$; 60° → $\approx 38 \text{ V}$; 30° → $\approx 48 \text{ V}$ (sim slightly lower).

3.2 Full-wave thyristor rectifier (I) — 2 SCR + 2 diodes



1. NEW → exp3_semi . ADD AC Voltage Source (169.7 , 60), Thyristor ×2 (T1, T2, Rs 1e7) , Diode ×2 (D1, D2, Rs 1e7) , Series RLC (RL 10 , 10e-3) , Voltage Measurement, Scope, Mean (120) , Display, powergui (Discrete 5e-6), 4 Terminators.
2. ADD Pulse Generator: Period 1/60 , Width 5 , Phase delay 1/60/6 ($\alpha = 60^\circ$).
3. ADD “Transport Delay” (Simulink Continuous) → Time delay 1/60/2 (= half cycle = 180°).

Node a = left-leg midpoint, b = right-leg midpoint, P = top rail, N = bottom rail.

4. Make T1, T2, D1, D2 vertical with the anode at the **bottom**.
5. WIRE T1 cathode → T2 cathode → RL top (this is rail **P**).
6. WIRE D1 anode → D2 anode → RL bottom (rail **N**).
7. WIRE T1 anode → D1 cathode (node **a**); T2 anode → D2 cathode (node **b**).
8. WIRE AC source: one terminal → node a, other terminal → node b.
9. WIRE Pulse Generator → T1 g. Branch the pulse wire → Transport Delay → T2 g.
10. m-ports → Terminators. Voltage Measurement across RL → Scope, → Mean(120) → Display.
11. RUN (0.1 s). Optional: T1 m → Demux → “SCR Current”, “SCR Voltage” scopes (for rating).

Two humps per cycle, each starting at $\alpha = 60^\circ$. $V_{avg} \approx (V_m/\pi)(1 + \cos \alpha) = 81 \text{ V}$ (sim a bit lower).

3.3 Full-wave thyristor rectifier (II) — 4 SCR

1. Save `exp3_semi` as `exp3_full`. Replace D1, D2 by Thyristors T4, T3 ($R_s = 1e7$), same position/orientation (anode at bottom).
2. ADD “Goto” tag **A** (input = Pulse Generator) and “Goto” tag **B** (input = Transport Delay output).
3. ADD four “From” blocks: **[A] → T1 g** (top-left) and **[A] → T3 g** (bottom-right); **[B] → T2 g** (top-right) and **[B] → T4 g** (bottom-left). Rule: the diagonal pair fires together.
4. RUN (0.1 s).

Same waveform as 3.2 for this small L. With very large L the output would go negative ($V_{avg} = (2V_m/\pi) \cos \alpha$).

3.4 Three-phase thyristor rectifier

$V_{LL} = 207.85 \text{ V rms}$ (120 V phase), 60 Hz, $R_L = 10 \Omega$, 10 mH, $\alpha = 0^\circ$ and 30° .

Block (ADD)	SET
Three-Phase Source	Yg, Phase-to-phase Vrms 207.8461, Freq 60, untick Internal
Three-Phase V-I Measurement	Voltage measurement phase-to-ground, Current measurement yes
Universal Bridge	Arms 3, $R_s = 1e7$, $C_s = \text{inf}$, device Thyristors , $R_{on} = 1e-3$, $L_{on} = 0$, $V_f = 0$
Series RLC Branch	$R_L = 10$, $10e-3$
PLL (3ph)	defaults (Min freq 45, Initial [0, 60], gains [180, 3200, 1], tick Enable automatic gain control)
Pulse Generator (Thyristor, 6-Pulse)	Generator type 6-pulse, Pulse width (deg) 60, Double pulsing unticked
Constant ×2	“alpha” = 30 (degrees); “Block” = 0
Voltage Measurement, Mean (360 = 6×60), Display, Scope, powergui (Discrete 5e-6)	

1. NEW → `exp3_3ph_scr`. ADD and SET all blocks.
2. WIRE Three-Phase Source A, B, C → V-I Measurement A, B, C; V-I a, b, c → Universal Bridge A, B, C.
3. WIRE Universal Bridge + → RL top; RL bottom → Universal Bridge −.
4. WIRE V-I Measurement **Vabc** output → PLL (3ph) input. PLL output **wt** → Pulse Generator (6-pulse) **wt** input. (Unused PLL output Freq → Terminator.)
5. WIRE Constant (alpha) → **alpha** input; Constant 0 → **Block** input.
6. WIRE Pulse Generator output **P** → Universal Bridge **g**.
7. Voltage Measurement across RL → Scope, → Mean(360) → Display. **Iabc** → Terminator.
8. RUN (0.1 s). Change alpha 0 → 30 → 60 and record.
9. (SCR current) Universal Bridge → Measurements **Device currents**; ADD Multimeter → select **Isw1: Universal Bridge** → Mean (60) → Display.

$\alpha = 0^\circ$: **280.6 V**; $\alpha = 30^\circ$: **242.5 V** (theory $1.35 V_{LL} \cos \alpha = 280.7 / 243.1$). Each SCR $I_{avg} = I_{load}/3$ ($\approx 9.36 \text{ A}$ at $\alpha = 0^\circ$).

3.5 Effect of a filter capacitor

1. NEW → `exp3_filter`. ADD AC Voltage Source (Peak 100, Freq 60), Universal Bridge (arms 2, device **Diodes**, $R_s = 1e7$, $C_s = \text{inf}$, $R_{on} = 1e-3$, $V_f = 0.8$, Measurements **Device currents**), Series RLC Branch $R = 10$, Voltage Measurement,

powergui (Discrete $1e-7$).

2. WIRE source terminals → bridge A and B. Bridge + → R top, R bottom → bridge -. Voltage Measurement across R.
3. v → Scope "Load Voltage", → Mean (120) → Display.
4. ADD Multimeter → select the first device current (Isw1 ... = diode D1) → Scope "D1 Current" and → Mean (60) → Display.
5. RUN (0.1 s). Note values (no capacitor).
6. ADD Series RLC Branch → Branch type **C**, Capacitance $1000e-6$, Measurements **Branch current**. WIRE it in parallel with R. RUN again.

No C: V_{avg} **62.06 V**, 0–98.4 V, i_{D1} peak < 10 A. With 1000 μ F: V_{avg} **79.17 V**, ripple 56.7–98.4 V, i_{D1} = narrow pulses, peak $\approx$ **39 A**. Capacitor current: average 0, peak 37.6 A, rms $\approx$ 12 A.

3.6 Coupled inductors

$V_{in} = 1$ V step across $L1 = 1$ mH; $L2 = 1.1$ mH in series with $R = 1 \Omega$; $M = 0.9$ mH.

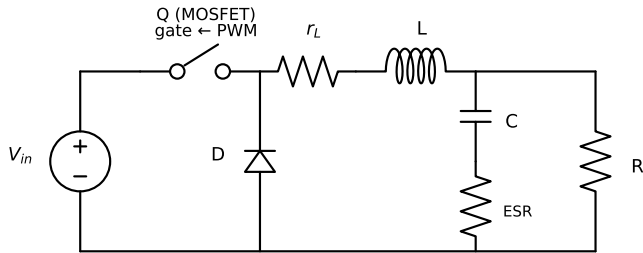
1. NEW → exp3_coupled . ADD "Mutual Inductance" → Type *Two or three windings with equal mutual terms* (or the default two-winding type), Winding 1 [R1 L1] [0 1e-3] , Winding 2 [R2 L2] [0 1.1e-3] , untick Three windings, Mutual [Rm Lm] [0 0.9e-3] , Measurements **Winding currents**.
2. ADD "Controlled Voltage Source" (Sources) + "Step" (Step time 0 , Initial value 0, Final value 1). WIRE Step → Controlled Voltage Source input.
3. WIRE Controlled Voltage Source + → winding 1 terminal 1 (dotted); source - → winding 1 terminal 2.
4. ADD Series RLC Branch R 1 . WIRE winding 2 terminal 1 (dotted) → R → winding 2 terminal 2.
5. ADD Multimeter → select both winding currents → Demux(2): out1 → Scope "IL1"; out2 → Gain -1 → Scope "IL2".
6. POWERGUI (Discrete $5e-6$ — or $1e-7$ for smoothness). RUN (Stop Time $2.5e-3$).

$k = M/\sqrt{L1L2} = 0.858$. IL2 rises exponentially to **0.9 A** ($= (M/L1) \cdot V/R$); IL1 keeps ramping (DC across an inductor).

Exp-04 — DC-DC Converters

4.1 Buck converter (open loop)

$V_{in} = 50 \text{ V}$, $f_s = 20 \text{ kHz}$, $D = 0.4$, $L = 400 \mu\text{H}$ ($r_L = 1 \text{ m}\Omega$), $C = 100 \mu\text{F}$ (ESR $50 \text{ m}\Omega$), $R = 20 \Omega$.



Block	SET
DC Voltage Source	Amplitude 50
Mosfet	defaults
Diode	Rs 1e7
Series RLC "Inductor"	RL: R 1e-3 , L 400e-6 , Measurements Branch current
Series RLC "Capacitor"	RC: R 50e-3 , C 100e-6
Series RLC "Load"	R 20
Pulse Generator	Amp 1 , Period 1/20e3 , Width 40 , Delay 0
powergui	Discrete 5e-7 (smoother than 5e-6)

1. NEW → exp4_buck . ADD and SET all blocks. Also ADD Voltage Measurement, Scope "Load Voltage", Mean (20e3), Display, 2 Terminators.
2. Place DC source vertical on the left (+ on top).
3. WIRE DC + → Mosfet **drain (d)**. Mosfet **source (s)** → node X.
4. Diode vertical, anode at the bottom: WIRE Diode cathode → node X; Diode anode → DC source - (this bottom wire is the ground rail).
5. WIRE node X → Inductor branch left; Inductor right → node Y.
6. Capacitor branch vertical: top → Y, bottom → ground rail. Load branch vertical: top → Y, bottom → ground rail.
7. WIRE Pulse Generator → Mosfet g. Mosfet m and Diode m → Terminators.
8. Voltage Measurement across Load → Scope, → Mean(20e3) → Display.
9. RUN (Stop Time 0.05). Zoom into steady state to read ripple (max - min).

$V_o \approx 20 \text{ V}$ ($= D \cdot V_{in}$, slightly lower), ripple $\approx 100 \text{ mV}$. Theory $\Delta V = (1-D)V_o / (8LCf_s^2) = 94 \text{ mV} + \text{ESR part}$.

4.2 Operating mode (CCM / DCM)

1. In exp4_buck (inductor Measurements = Branch current). ADD Multimeter → select Ib: Inductor → Scope "Inductor Current".
2. RUN (0.05 s). Zoom into the last few switching periods. If the minimum is **above zero** → CCM.
3. Change Load to 40 → RUN. Current touches 0 and stays 0 for part of the period → DCM.

$L_{crit} = (1-D)R / (2f_s)$: $R = 20 \Omega \rightarrow 0.3 \text{ mH} < 0.4 \text{ mH} \rightarrow \text{CCM}$; $R = 40 \Omega \rightarrow 0.6 \text{ mH} > 0.4 \text{ mH} \rightarrow \text{DCM}$. Boundary $R = 26.7 \Omega$.

4.3 Effect of input voltage change

1. Save as exp4_vin_step . Delete the DC Voltage Source. ADD "Controlled Voltage Source" in the same place (+ on top) and "Step" → Step time 50e-3 , Initial value 50 , Final value 40 .
2. WIRE Step → Controlled Voltage Source input. Load back to 20 .

3. RUN (Stop Time 0.1).
4. Repeat with Final value 60.

Open loop: V_o follows $D \cdot V_{in}$; 20 V $\rightarrow$ 16 V (or $\rightarrow$ 24 V) after 50 ms with an oscillatory transient.

4.4 Effect of output load change

(a) Load decrease 20 Ω $\rightarrow$ 10 Ω (add a parallel resistor)

1. Save exp4_buck as exp4_load_step. ADD Series RLC R 20 ("R2") and a second Mosfet ("S2") + Step (time 50e-3, initial 0, final 1).
2. WIRE node Y $\rightarrow$ R2 top; R2 bottom $\rightarrow$ S2 drain; S2 source $\rightarrow$ ground rail; Step $\rightarrow$ S2 g; S2 m $\rightarrow$ Terminator.
3. RUN (0.1 s): for $t > 50$ ms load = 20||20 = 10 Ω $\rightarrow$ small change in V_o .

(b) Load increase 20 Ω $\rightarrow$ 40 Ω (series resistor)

1. Put R1 = 20 Ω and R2 = 20 Ω in **series** between Y and ground; connect S2 drain to the middle point, S2 source to ground (S2 shorts R2). Step: initial 1, final 0, time 50e-3.
2. RUN: after 50 ms load = 40 Ω $\rightarrow$ V_o rises (converter enters DCM).

4.5 Generation of PWM signals (carrier comparison)

1. NEW $\rightarrow$ exp4_pwm. ADD "Sawtooth Generator" (Specialized Power Systems $\rightarrow$ Control) $\rightarrow$ Frequency 20e3, Phase 0. (Output goes -1 $\rightarrow$ +1.)
2. ADD Constant 1 and "Sum" (list of signs ++): sawtooth + 1. Then ADD "Gain" 0.5 $\rightarrow$ carrier now 0 $\rightarrow$ 1.
3. ADD Constant "Duty cycle" = 0.4 $\rightarrow$ "Saturation" (Upper limit 1, Lower limit 0).
4. ADD "Relational Operator" $\rightarrow$ Relational operator $\geq$. WIRE Saturation $\rightarrow$ **top** input; carrier (Gain output) $\rightarrow$ **bottom** input.
5. Relational Operator output $\rightarrow$ Scope "PWM". Carrier $\rightarrow$ Scope "Ramp Signal".
6. For the complement: ADD "Logical Operator" $\rightarrow$ Operator **NOT**; Relational output $\rightarrow$ NOT $\rightarrow$ Scope "Inverted PWM".
7. No electrical blocks here, so powergui is not required (if MATLAB asks for one, add powergui $\rightarrow$ Continuous). Simulation tab $\rightarrow$ Stop Time 5e-4 $\rightarrow$ RUN. Check with cursors: period 50 μ s, ON 20 μ s.

$D = V_{ref}/V_{carrier,peak} = 0.4 \rightarrow 40\%$; NOT $\rightarrow 60\%$; frequency = carrier frequency.

4.6 PWM Generator (DC-DC) block

1. NEW $\rightarrow$ exp4_pwmgen. Build the buck of 4.1 with: DC 10 V, Inductor RL 0.1 Ω / 100e-6, Capacitor RC 0.1 Ω / 100e-6, Load 5 Ω .
2. Instead of Pulse Generator: ADD Constant 0.4 $\rightarrow$ Saturation (0, 1) $\rightarrow$ "PWM Generator (DC-DC)" (Switching frequency 100e3) $\rightarrow$ Mosfet g.
3. POWERGUI: Simulation type **Continuous**.
4. Voltage Measurement across load $\rightarrow$ Scope, $\rightarrow$ Mean (100e3) $\rightarrow$ Display. PWM output $\rightarrow$ Scope "PWM".
5. RUN (0.02 s).

V_o 3.411–3.436 V, ripple $\approx$ 25 mV, $V_{avg} \approx$ 3.42 V (ideal 4 V; less due to diode 0.8 V drop and r_L).

4.7 Closed-loop control of a buck converter (PI)

1. Save exp4_pwmgen as exp4_closed. Delete the Constant 0.4.
2. ADD "Step" (reference): Step time 50e-3, Initial 3, Final 6.
3. ADD "Sum" $\rightarrow$ List of signs +- . WIRE Step $\rightarrow$ "+" input; measured V_o (Mean output) $\rightarrow$ "-" input.
4. ADD "PID Controller" $\rightarrow$ Controller **PI**, Proportional (P) 1e-3, Integral (I) 100.
5. WIRE Sum $\rightarrow$ PID $\rightarrow$ Saturation (0, 1) $\rightarrow$ PWM Generator (DC-DC) $\rightarrow$ Mosfet g. PID output $\rightarrow$ Scope "Duty Cycle".

6. RUN (Stop Time 0.1).

V_o tracks 3 V, then 6 V after 50 ms, zero steady-state error; duty cycle increases at 50 ms. Test: change V_{in} — output stays at reference.

4.8 Flyback converter

$24\text{ V} \rightarrow 5\text{ V}$, $f_s = 40\text{ kHz}$, $D = 0.54$, $C = 470\text{ }\mu\text{F}$ (ESR 50 m Ω), $R = 5\text{ }\Omega$, $L_m = 500\text{ }\mu\text{H}$.

1. NEW $\rightarrow$ exp4_flyback . ADD DC Voltage Source 24 , Mosfet, Diode (Rs 1e7), Series RLC RC (50e-3 , 470e-6), Series RLC R 5 , powergui **Continuous**.
2. ADD “Linear Transformer” $\rightarrow$ Units SI, Nominal power and frequency [5 40e3] , Winding 1 [24 0 0] , Winding 2 [5 0 0] , untick Three windings, Magnetization [Rm Lm] [1e7 500e-6] .
3. Primary: WIRE DC + $\rightarrow$ winding 1 **top** (dot); winding 1 bottom $\rightarrow$ Mosfet drain; Mosfet source $\rightarrow$ DC -.
4. Secondary (flyback = opposite polarity): winding 2 **bottom** $\rightarrow$ Diode anode; Diode cathode $\rightarrow$ output + node; winding 2 **top** (dot) $\rightarrow$ output - node.
5. Capacitor branch and Load between output + and output -.
6. Constant 0.54 $\rightarrow$ PWM Generator (DC-DC) (40e3) $\rightarrow$ Mosfet g.
7. Voltage Measurements: across load $\rightarrow$ Scope “Load Voltage”; across Mosfet drain-source $\rightarrow$ Scope “V MOSFET”.
8. RUN (Stop Time 0.05).

$V_o \approx 5\text{ V}$. V_{DS} spike $\approx 65\text{ V}$ at start, steady $\approx 52\text{ V}$ ($= V_{in} + (N1/N2)(V_o + V_F)$) $\rightarrow$ MOSFET must withstand it. If V_o is far from 5 V, swap the two secondary terminals (polarity).

4.9 Efficiency of the flyback

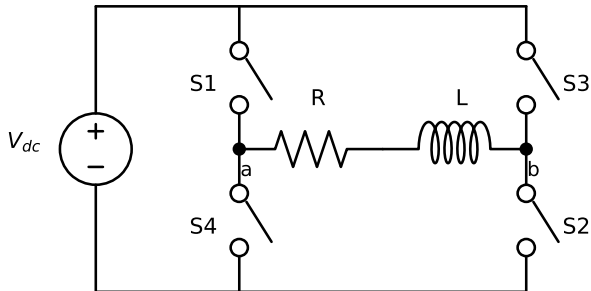
1. Save as exp4_eff . DC source $\rightarrow$ Measurements **Voltage**.
2. ADD Series RLC R 1e-6 (“Sense Resistor”, Measurements **Branch current**) in series with the DC source + lead.
3. Load $\rightarrow$ Measurements **Branch voltage and current**.
4. Multimeter 1: select DC source voltage + Sense Resistor current $\rightarrow$ Demux(2) $\rightarrow$ Product $\rightarrow$ Mean (40e3) = P_{in} .
5. Multimeter 2: select load U_b and I_b $\rightarrow$ Demux(2) $\rightarrow$ Product $\rightarrow$ Mean (40e3) = P_{out} .
6. Divide $P_{out} \div P_{in} \rightarrow$ Gain 100 $\rightarrow$ Display “Efficiency (%)”. RUN (0.05 s).

$\eta \approx 84.45\%$. Remaining 15.55% is lost mostly in the diode ($V_F \times I$) and MOSFET R_{on} .

Exp-05 — Inverters

5.1 Single-phase full-bridge unipolar PWM inverter

$V_{dc} = 100 \text{ V}$, RL load $25 \Omega + 1 \text{ mH}$, output 50 Hz , $m_a = 0.7$, $m_f = 23$ ($f_s = 1150 \text{ Hz}$).



S1/S4 = left leg (point a), S3/S2 = right leg (point b). Load between a and b.

Block	SET
DC Voltage Source	100
Mosfet ×4 (S1, S4, S3, S2)	defaults
Series RLC "Load"	RL 25 , 1e-3 , Measurements Branch voltage and current
Repeating Sequence "Carrier"	Time values [0 1 2 3 4]*1/(23*50*4) , Output values [0 1 0 -1 0]
Sine Wave 1 / Sine Wave 2	Amplitude 0.7 , Frequency 2*pi*50 (rad/s), Phase 0 / pi
Relational Operator ×2	>=
Logical Operator ×2	NOT
powergui	Discrete 5e-6

- NEW → exp5_1ph_inv . ADD and SET all blocks.
- DC source vertical. WIRE DC + → top rail; DC - → bottom rail.
- Left leg: S1 drain → top rail, S1 source → point a; S4 drain → point a, S4 source → bottom rail.
- Right leg: S3 drain → top rail, S3 source → point b; S2 drain → point b, S2 source → bottom rail.
- Load: left terminal → a, right terminal → b.
- Sine Wave 1 → Relational Operator 1 top; Carrier → Relational Operator 1 bottom. Output = signal A.
- Sine Wave 2 → Relational Operator 2 top; Carrier → bottom. Output = signal B.
- Gates: A → S1 g; A → NOT → S4 g; B → S3 g; B → NOT → S2 g. (Use Goto/From [A], [B] to keep it tidy.) All m-ports → Terminators.
- Voltage Measurement across the load (+ at a) → Scope "Load Voltage". Current Measurement in series with load (or Multimeter Ib) → Scope "Load Current".
- RUN (Stop Time 0.1).
- Averaged voltage: v → Mean (frequency 23*50) → Scope "Average Load Voltage".
- RMS: v → RMS (50) → Display. Fundamental: v → Fourier (50 , Harmonic n 1) → Display (magnitude).

Load voltage = pulses at +100, 0, -100 V. Averaged voltage = sine of amplitude **68.48 V**; V_{rms} **66.2 V**; fundamental **69.32 V** peak (theory $m_a V_{dc} = 70 \text{ V}$).

If you get a data-type error at a Mosfet gate: open the Relational/Logical Operator → Signal Attributes → Output data type = double , or put a "Data Type Conversion" block (Output data type double) before the gate.

5.2 THD block

1. In `exp5_1ph_inv` : ADD “THD” (Sensors and Measurements) → Fundamental frequency `50` . `v` → THD → Display “Voltage THD”.
2. ADD a second THD (`50`): load current → THD → Display “Current THD”.
3. RUN (0.1 s). Change m_a (both sines’ Amplitude) to 0.5, 0.9, 1.0 and record THD.

$m_a = 0.7$: V-THD **90.78%**, I-THD **69.4%** (inductor filters high frequencies). Check: $\text{THD} = \sqrt{(V_{\text{rms}}^2 - V_{1,\text{rms}}^2)/V_{1,\text{rms}}^2} = \sqrt{(66.2^2 - 49.02^2)/49.02^2} = 0.907 \checkmark$.

5.3 Harmonic analysis with the FFT Analyzer

1. Double-click Scope “Load Voltage” → Settings (gear / Configuration Properties) → **Logging** tab → tick **Log data to workspace** → Variable name `LoadVoltage` → Save format **Structure With Time** → OK.
2. RUN. Variable `out` appears in the Workspace.
3. In the command window type: `LoadVoltage = out.LoadVoltage;` → Enter.
4. Double-click powergui → **Tools** tab → **FFT Analysis**.
5. In FFT Analyzer: select the signal `LoadVoltage`, Start time `0.08` , Number of cycles `1` , Fundamental frequency `50` , Max frequency `1250` → **Compute FFT**. Read THD and bars.
6. Set Max frequency `2500` → Compute again → see dominant harmonics.
7. Display style **List (relative to specific base)**, Base value `1` → shows peak volts of each harmonic.

Fundamental 69.32 V; 3rd harmonic 0.15% (0.10 V); dominant harmonics **43, 45, 47, 49** (around $2m_f = 46$ — typical for unipolar PWM).

5.4 Three-phase PWM inverter

$V_{dc} = 100$ V, Y-connected RL 25 Ω + 1 mH per phase, 50 Hz, $m_a = 0.7$, $f_s = 1150$ Hz.

1. NEW → `exp5_3ph_inv` . ADD DC Voltage Source `100` , 6 Mosfets, 3 Series RLC branches (RL `25` , `1e-3` , Measurements Branch voltage and current, names phase A, B, C), powergui (Discrete 5e-6).
2. Three legs exactly like one leg of 5.1: upper Mosfet drain → + rail, source → midpoint; lower Mosfet drain → midpoint, source → - rail. Midpoints = **A, B, C**.
3. WIRE A → phase A branch left, B → phase B left, C → phase C left; WIRE the three right terminals together = point **N**.
4. PWM: one Repeating Sequence carrier (same settings as 5.1). Three Sine Waves, all Amplitude `0.7` , Frequency `2*pi*50` , Phase `0` , `2*pi/3` , `-2*pi/3` .
5. Three Relational Operators ($\geq$): sine k → top, carrier → bottom → Goto [A], [B], [C].
6. From [A] → upper-A gate; From [A] → NOT → lower-A gate. Same for B and C. m-ports → Terminators.
7. Multimeter → select Ub: phase A, Ub: phase B → Demux(2) → Scope “V_AN”. For line voltage: ADD Sum (`+` - `-`): out1 - out2 → Scope “Line-Line Voltage”.
8. RUN (0.1 s). Measure RMS/THD with the blocks from 5.1–5.2 (frequency 50).

V_{AN} : multi-level stepped PWM; V_{AB} : pulses ± 100 V. Theory: $V_{AN,1,\text{peak}} = m_a V_{dc}/2 = 35$ V; $V_{LL,1,\text{rms}} = 0.612 m_a V_{dc} = 42.8$ V.

5.5 Connection Port block (subsystem with electrical terminals)

1. Save as `exp5_connport` . Delete the wires from the midpoints A, B, C to the load.
2. ADD three “Connection Port” blocks (Simscape → Utilities, or type “Connection Port”) → Port location on parent subsystem **Right**. WIRE them to midpoints A, B, C.
3. Drag a rectangle around the power stage (DC source, 6 Mosfets, From/NOT blocks, 3 Connection Ports) → `Ctrl+G` → name “Inverter”.
4. Back on top level, WIRE the subsystem’s three right terminals to the load branches.

5. Open the PWM section's Goto blocks → **Tag visibility = global** (the From blocks are now inside the subsystem).
Optionally make the PWM section another subsystem.
6. RUN — results identical to 5.4.

C. If they give a circuit that is not in the manual

Circuit	How to build (same blocks)	Theory
1- ϕ full-wave diode bridge	Universal Bridge 2 arms Diodes (or 4 diodes like 3.3 with diodes); Mean @ 2f	$V_{avg} = 2V_m/\pi$, $V_{rms} = V_m/\sqrt{2}$
Center-tap full-wave	Linear Transformer with 3 windings (tick Three windings), W2 and W3 in series, centre → load bottom, two diodes → load top	$2V_m/\pi$
Boost	$V_{in} + \rightarrow$ Inductor $\rightarrow$ node X; Mosfet drain X, source $\rightarrow$ ground; Diode anode X, cathode $\rightarrow V_o$; C R from V_o to ground	$V_o = V_{in}/(1-D)$
Buck-boost	$V_{in} + \rightarrow$ Mosfet d; Mosfet s $\rightarrow$ X; Inductor X $\rightarrow$ ground; Diode cathode X, anode $\rightarrow V_o$ (negative); C R from V_o to ground	$V_o = -D \cdot V_{in}/(1-D)$
Bipolar SPWM inverter	One sine vs carrier $\rightarrow$ A; A $\rightarrow$ S1 and S2; NOT A $\rightarrow$ S3 and S4	$V_{1,pk} = m_a V_{dc}$; harmonics at m_f
Square-wave inverter	Pulse Generator (Period 1/f, Width 50) $\rightarrow$ S1, S2; NOT $\rightarrow$ S3, S4	$V_{1,pk} = 4V_{dc}/\pi$
AC voltage controller	Two antiparallel Thyristors in series with the load; T1 delay $\alpha/(360f)$, T2 via Transport Delay 1/(2f)	$V_{rms} = V_s \sqrt{[(1/\pi)(\pi - \alpha + \sin 2\alpha/2)]}$ (R load)

Troubleshooting (read the red error, then): “must contain a powergui” → add powergui · lines won't connect → wrong library block (take Specialized Power Systems) · SCR output zero → check Pulse Amplitude 1, Period 1/f, wire to g, not to m · Mean shows nonsense → wrong fundamental frequency or stop time too short · waveform jagged → smaller powergui step (1e-7) or Continuous · algebraic-loop/unconnected warnings → Terminators · From block red → Goto tag visibility global · simulation extremely slow → shorten stop time or increase step to 5e-6.

Source: PE_EXP_01–05 (F. Asadi, Simulation of Power Electronics Circuits with MATLAB/Simulink, Ch. 3–7). Menu names follow R2024b; if a menu looks slightly different on your PC, the block settings/values stay the same.